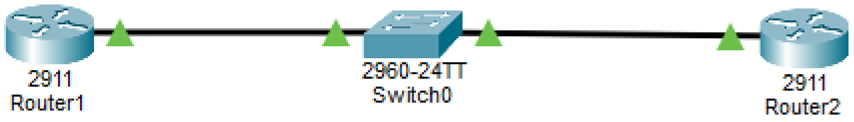
Topology



Hostname

```
Router(config-if)#hostname R1
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router(config-if)#hostname R2
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch(config)#hostname SW1
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

Ip Address configurations

R1

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#no shutdown
```

R2

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.10.10.2 255.0.0.0
Router(config-if)#ip address 10.10.10.2 255.0.0.0
Router(config-if)#no shutdown
```

SW1

```
SW1#conf t
Enter configuration commands, one per line.  End with
SW1(config)#int vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown
```

Ping from SW1 to R2

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/30/82 ms
```

36. Verifying Neighbors

```
R1#show cdp
Global CDP information:
    Sending CDP packets every 60 seconds
    Sending a holdtime value of 180 seconds
    Sending CDPv2 advertisements is enabled
```

37. Prevent R1 from discovering SW1 via CDP

```
R1(config)#int g0/0
R1(config-if)#no cdp enable
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show cdp
Global CDP information:
    Sending CDP packets every 60 seconds
    Sending a holdtime value of 180 seconds
    Sending CDPv2 advertisements is enabled
R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
SW1          Gig 0/0          99            S        2960        Fas 0/1
```

38. Flush the CDP cache on R1

```
R1#conf t
Enter configuration cc
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#end
```

39.

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID     Local Intrfce    Holdtme    Capability   Platform    Port ID
```

40.

```
SW1>ena
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
```

41.

```
SW1(config)#int fa0/2
SW1(config-if)#shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively dow

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual administratively down down
```

42.

```
SW1(config)#int fa0/2
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state t

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
```

43.

```
;W1>ena
;W1#conf t
Inter configuration commands, one per line.  End with CNTL/Z.
;W1(config)#int fa0/2
;W1(config-if)#duplex half
;W1(config-if)#end
;W1#
;SYS-5-CONFIG_I: Configured from console by console

;W1#show interfaces fa0/2
'astEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.ffc2.ea02 (bia 00d0.ffc2.
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output
```

44.

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#int fa0/2
SW1(config-if)#duplex full
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interfaces fa0/2
                      ^
% Invalid input detected at '^' marker.

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.ffc2.ea02 (bia 00d0.ffc2.ea
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output d
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
```

45.

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#int fa0/2
SW1(config-if)#speed 10
SW1(config-if)#end

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.ffc2.ea02 (bia 00d0
 BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
```

46.

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
```

47. A duplex mismatch causes collisions, packet loss and slow performance because the two devices are using different duplex settings.